RESEARCH PRESENTATION

Does every query need every retriever?

From Blended RAG to a learned cascade that spends neural computation only when it is likely to help.

A study of retrieval quality, recall budgets, and execution cost.

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Completed experimental scope · 21 October 2026. Based on the project manuscript and verified result reports.

> **Say:** We started with the idea that some questions can use BM25 alone. Our project studies when that shortcut helps, how much quality it costs, and why a fast first stage is essential.

THE STARTING POINT

Blended RAG improves the evidence given to the generator.

Sawarkar, Mangal and Solanki (2024) study semantic indexes and hybrid queries across document fields.

Keyword search

BM25 finds exact words, names, and technical terms.

Dense search

Vector similarity retrieves passages with related meaning.

Learned sparse search

ELSER weights related vocabulary to support semantic matching.

The paper explores index and field-query combinations, including Best Fields, to improve retrieval and generated answers.

Source: Blended RAG, arXiv:2404.07220v2, sections III–V. Our three-way RRF baseline is an adaptation of this broader approach.

> **Say:** The paper's central contribution concerns retrieval quality through semantic indexes and hybrid field queries. Avoid describing the original paper as exactly our BM25, MiniLM, SPLADE, and RRF code.

THE GAP WE INVESTIGATE

A stronger retrieval recipe still leaves a cost question.

“Does this particular question need neural retrieval?”

The paper evaluates retrieval and answer quality across query strategies. It does not establish a learned BM25-first gate with a recall budget for each query.

Our static adaptation executes BM25, Dense, and SPLADE for every query. Easy lexical matches may need less work.

The trade-off

Skipping a neural branch can save time, but BM25 may miss documents that semantic retrieval finds.

Our question: how much mean retrieval time can we save while retaining an explicit share of fusion recall?

Quality and latency must be measured together. A high early-exit count alone is insufficient.

Paper context: Blended RAG. Efficiency question and static baseline: project manuscript, sections 1–2.

> **Say:** This is the gap our study addresses, rather than a claim that the original system is universally slow. We measure the cost in our implementation.

OUR IMPLEMENTATION

We built a Python baseline, then made execution conditional.

Part	Original paper	Our project
Search platform	Elasticsearch	Python + FAISS + sparse matrices
Neural retrieval	Dense index + ELSER	MiniLM + SPLADE
Hybrid strategy	Index and field-query combinations	Best Fields BM25 + three-way RRF
New execution policy	Outside our matched evaluation	Indexed BM25 + a Ridge gate

All reported speedups compare versions of our own pipeline.

Project manuscript, sections 3–4. Published ELSER quality and our measured MiniLM/SPLADE quality use different systems and protocols.

> **Say:** We reuse pretrained MiniLM and SPLADE. The learned component we train is the small Ridge gate. Testing our changes directly in Elasticsearch and ELSER remains future work.

QUERY EXECUTION

BM25 runs first. The gate decides what comes next.

Question
Incoming query

→

Indexed BM25
Top 100 + 11 cheap features

→

Ridge gate
Predict the recall gain of fusion

Small predicted gain

Prediction ≤ saved threshold

Return BM25's ranking. Dense and SPLADE receive no query call.

Larger predicted gain

Prediction > saved threshold

Run Dense and SPLADE. Fuse both rankings with the existing BM25 ranking using RRF.

Runtime: src/retention_tradeoff.py. RRF adds 1 / (60 + rank) from each retriever and returns the fused top 100.

> **Say:** The gate only uses query and BM25 information. It does not need the correct documents or neural rankings to decide. Both neural retrievers search the corpus when escalation occurs.

HOW THE GATE LEARNS

Training measures the gain. New queries use a prediction.

01 Run BM25 and fusion on 2,590 NFCorpus training queries.

Known relevance labels let us calculate both recalls.

02 Train standardized Ridge on 11 features.

Target gain = max(0, fusion recall − BM25 recall).

03 Sort out-of-fold predictions from small to large.

Add the actual losses of proposed early exits.

04 Save the largest cutoff inside the loss budget.

Fit the final Ridge model on all training queries.

Real r95 calibration

Target: keep at least 95% of fusion's mean Recall@100.

Allowed total loss
= 0.05 × 823.0387
≈ 41.15

Selected prediction threshold: **0.0756657**
Early exits: **604 / 2,590**
Calibration recall retained: **95.42%**

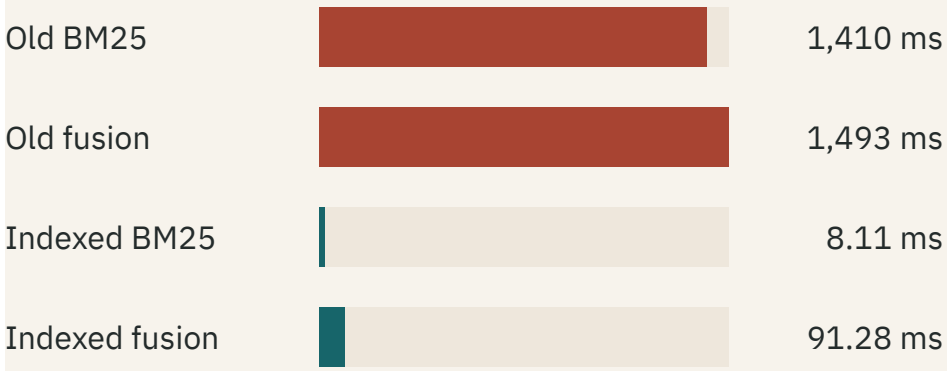
Experiments 12 and 16; saved NFCorpus Ridge11 policy. r90/r92/r95/r99 name training targets. They are neither probabilities nor guarantees on new datasets.

> **Say:** The budget applies to all selected training queries together. The threshold applies to each query separately. At test time, the gate compares the predicted gain with the fixed threshold; there is no live accumulating loss budget.

THE EXECUTION BOTTLENECK

A cascade cannot avoid the first-stage cost.

Our original TREC pipeline spent most of its query time in BM25. Skipping later neural calls could save only a small share.



What we changed

Precompute term contributions in postings and retrieve the top scores with a partial sort.

We checked score and ranking agreement, then gave the same optimization to **both fusion and cascade**.

Indexing reduces the unavoidable cost. Routing then saves part of the remaining work.

TREC means from separate original/indexed runs. Warm retrieval only. Experiments 24, 25, and 30 verify scorer compatibility and saved-ranking agreement.

> **Say:** The indexing change preserves measured retrieval outputs. Its speedup and the gate's speedup are distinct contributions. The old-versus-new ratios also include variation between runs.

EVALUATION PROTOCOL

Freeze the policy. Measure the calls that actually run.

01 **Train and calibrate on NFCorpus train only.**
Inspect NFCorpus dev as development evidence.

02 **Transfer frozen weights and thresholds.**
SCIDOCS and paper-dataset evaluations do not retrain the gate.

03 **Measure complete query paths and branch counts.**
Warm components, synchronize GPU timing, and randomize method order.

04 **Score saved rankings independently.**
Compare all gates with fusion and each standalone retriever.

Final indexed scope

50

TREC-COVID queries · 171,332 documents

256

Fixed SQuAD questions · 20,958 documents

Three completed warmed timing passes for each indexed study. The original full SQuAD run covers 10,570 questions with one completed pass.

Experiments 17, 18, 20, 25, 29–31. Indexed TREC follows already-inspected quality evidence; it is an engineering follow-up. Timing excludes indexing and answer generation.

> **Say:** These are actual conditional calls, unlike our early notebook simulation. The indexed SQuAD result applies only to the fixed sample. The tests retain full document corpora.

THE MEASURED TRADE-OFF

More skipping buys speed

More skipping buys speed at a possible recall cost.

TREC-COVID: mean retrieval time; retention relative to indexed fusion.

Method	Mean time	Recall retained	Speedup vs indexed fusion
Indexed fusion	91.28 ms	100.00%	1.00×
r99 · primary	73.90 ms	96.68%	1.24×
r95 · secondary	55.66 ms	91.95%	1.64×
r90 · secondary	40.41 ms	84.53%	2.26×

TREC uncertainty: r99 retention interval 93.58–98.75%. r95 interval 86.88–95.76%, crossing the 90% floor. r90 fails that floor.

SQuAD sample: fastest gate 18.59 ms versus fusion 29.55 ms. **1.59× speedup at 99.22% retention.**

The primary TREC gain is in mean time. Its p95 is slightly higher than fusion's.

docs/FINAL_RESULTS.md and docs/SPEEDUP_SUMMARY.md. Retention = mean Recall@100 of method / mean Recall@100 of fusion. Nominal targets differ from observed retention.

> **Say:** The primary TREC result saves 19.04% mean time against equally indexed fusion. Its 20.21× ratio against our original fusion includes indexing as well as routing and compares separate runs. The 99 in r99 describes its training target, not the measured test retention.

CONCLUSION

Conditional retrieval helps when the first stage is cheap enough.

We demonstrated a measured execution improvement over our full-fusion baseline.

What we established

Real neural-call skipping, a compatible indexed BM25 scorer, and an explicit quality–latency curve with frozen learned policies.

What limits the claim

SPLADE alone is faster with higher recall than the primary TREC cascade. On SQuAD, BM25 alone already clears a 90% retention floor.

Next research step: test the approach in the original Elasticsearch/ELSER system, compare modern adaptive baselines, and evaluate generated answers. Full current NQ and HotpotQA remain pending.

Read the evidence: project manuscript · final results · all operating points · original paper

> **Say:** Our contribution is an empirical engineering study. We support additional mean-time savings against matched fusion, while reporting stronger standalone choices and failed retention targets. These results do not establish preserved answer quality or superiority over the published Blended RAG system.